What Changes Enzyme Activity?

Biology 2e chapter 6 reading connection | Teacher guide / classroom preparation

Learning target

How does temperature affect the initial rate of gas collection from potato catalase?

Timing and materials

One class period after a teacher pilot.

Potato extract, teacher-diluted 1% hydrogen peroxide, water baths, thermometer, 100 mL flask, fitted delivery tube, 50 mL gas syringe with free plunger, graduated pipettes, timer, eye protection.

Classroom route

1. Teacher pilot: prepare one well-mixed batch of filtered potato extract. Try 2 mL extract and 5 mL 1% peroxide, with identical apparatus and quantities in every trial. Reduce extract if the syringe approaches capacity. These starting quantities need local validation.

2. Equilibrate separate reagents at 10, 22 or 35 C. Add peroxide last, promptly connect the gas collector, and start timing consistently. A teacher-approved addition port reduces gas lost during setup. Never obstruct the plunger or pressurize a sealed vessel.

3. Read collected gas every 30 seconds for 120 seconds. Use three fresh mixtures per temperature. Include an optional water-instead-of-extract control; avoid skin and eye contact with peroxide.

Pilot the physical procedure with your own equipment. Follow school handling and disposal rules and the purchased kit instructions. The virtual numerical settings are not an equipment prescription.

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Hybrid or substitute assignment

Assign all virtual, all physical, or specified samples in each mode. The classroom entry button is available during collection. Students complete the same prediction and evidence questions. Record classroom time series on the worksheet; the online classroom form stores endpoint measurements and notes. Keep sources and measurement scales separate.

Guided virtual sequence

Students prepare each sample, inspect the apparatus or classify specimens, then explicitly collect data. All assigned trials are required before exploration. Reload pauses the clock and restores saved work. Downloads retain source labels; no automatic grade is submitted.

Expected reasoning

In the modeled 10-35 C comparison, warmer conditions increase initial gas collection rate. This limited range does not establish the temperature optimum or demonstrate denaturation. Check equal extract concentration and gas leaks before interpreting physical differences.

Feedback and scoring

Criterion	Points	Evidence
Prediction and controls	0-2	Testable claim; comparable samples and controlled variables.
Observations	0-2	Complete, labeled records with units and source.
Interpretation	0-2	Correct calculations or categorical comparisons; claim tied to evidence.
Limitations	0-2	Specific uncertainty and a useful follow-up investigation.

Score reasoning and data quality, not agreement of physical results with the simulation. Manual tallies are student observations; a matching total does not prove every category is correct. Do not replace student counts with the model answer. The optional Sordaria extension retains classification feedback.

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Model limitations

Illustrative temperature response with substrate limitation; not a fit to a potato dataset. Gas dissolution, leaks, mixing delay and thermal expansion are omitted. The 10-40 C range does not model heat denaturation.

Troubleshooting

Confirm the preparation steps are finished before starting or collecting. Corn and fly activities require manual tallies summing to the sample size plus review; optional specimen marks do not classify or tally automatically. Chromosome models require all four gametes plus a reviewed tally; cell measurements require a focused 40x view; gels require staining and three size estimates; fluorescence labs require the viewer. Use the visible Collect data and continue button to advance.

For physical work, preserve unexpected observations, check the control and instrument setup, then repeat with fresh material where appropriate. Never replace measured results with the virtual answer.

Evidence and optional classroom resources

Enzyme reaction investigation

https://www.sciencebuddies.org/science-fair-projects/project-ideas/BioChem_p011/biotechnology-techniques/enzyme-catalyzed-reactions-what-affects-their-rates

External resources remain on their publishers' websites. This packet contains original text and tables, not supplier worksheets. Chapter numbers are reading connections. No supplier or textbook publisher endorsement is implied.

Teacher pilot checklist

Confirm availability and preparation time; test controls; check units and observation windows; verify handling/disposal; choose class grouping and accessible alternatives; keep notes for revision.